

Module 2: Linear Algebra

Vectors, Transformations, Spaces & Structures — Exam Reference Notes

Mathematics for Computing

September 22, 2026

Roadmap for Module 2

- ① Algebraic Structures (Groups, Rings, Fields)
- ② Vectors & Coordinate Systems
- ③ Vector Addition
- ④ Scalar Multiplication
- ⑤ Linear Combinations, Span, Basis
- ⑥ Matrices as Transformations
- ⑦ 3D Linear Transformations
- ⑧ The Determinant
- ⑨ Inverse, Column Space, Null Space
- ⑩ Nonsquare Matrices Between Dimensions
- ⑪ Dot Product & Duality
- ⑫ Hyperplanes
- ⑬ Cross Product
- ⑭ Cross Product via Transformations
- ⑮ Cramer's Rule
- ⑯ Change of Basis
- ⑰ Eigenvectors & Eigenvalues
- ⑱ Vector Spaces
- ⑲ Mathematical Spaces

Definition: Algebraic Structure.

An algebraic structure is a pair $(S, *)$ where S is a non-empty set and $* : S \times S \rightarrow S$ is a binary operation on S (i.e., combining two elements of S always gives another element of S — this is called *closure*).

Key Idea to Remember: Every "number system" you use in linear algebra (integers, reals, vectors, matrices) is an algebraic structure whose behaviour is fixed by a short list of axioms. Learn the axioms once, and you understand every structure built on top of them.

Groups

Definition: Group.

A set G with operation $*$ is a **group** $(G, *)$ if:

- ① **Closure:** $a * b \in G$ for all $a, b \in G$
- ② **Associativity:** $(a * b) * c = a * (b * c)$
- ③ **Identity:** $\exists e \in G$ such that $a * e = e * a = a$
- ④ **Inverse:** $\forall a \in G, \exists a^{-1} \in G$ such that $a * a^{-1} = e$

If also $a * b = b * a$ for all a, b , the group is **abelian** (commutative).

Example.

$(\mathbb{R}, +)$ is an abelian group: identity is 0, inverse of a is $-a$. $(\mathbb{R}^n, +)$ — ordinary vector addition — is also an abelian group: this is exactly why "vectors can be added and undone" always works.

In Plain Words: Think of a group like a locked room with a return ticket: whatever move you make $(*)$, you can always find a move that brings you back to where you started (inverse), and there is always a "do nothing" move (identity).

Rings and Fields

Definition: Ring.

A **ring** $(R, +, \cdot)$ is a set with two operations such that $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$: $a \cdot (b + c) = a \cdot b + a \cdot c$.

Definition: Field.

A **field** $(F, +, \cdot)$ is a ring in which $(F \setminus \{0\}, \cdot)$ is also an abelian group (every non-zero element has a multiplicative inverse). Example: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$.

Key Idea to Remember: A **vector space** (Slide 18) is always defined "over a field" — the scalars you are allowed to multiply vectors by must come from a field, because you need to add, subtract, multiply, and divide those scalars safely. This is the direct link between Section 1 and everything that follows.

Vectors and Coordinate Systems

Definition.

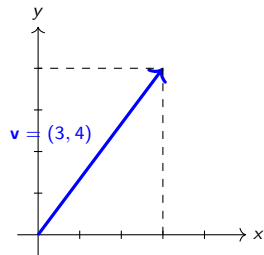
A **vector** in $\mathbb{R}^n$ is an ordered n -tuple of real numbers, written as a column:
 $\mathbf{v} = (v_1, v_2, \dots, v_n)$. Geometrically it is an arrow starting at the origin of a **coordinate system** and ending at the point $(v_1, \dots, v_n)$.

In Plain Words: Imagine standing at a crossroads (the origin). A vector is an instruction: "walk 3 steps East, then 4 steps North."

Example.

$\mathbf{v} = (3, 4)$: move 3 units along x , 4 units along y . Length
 $\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5$.

Key Idea to Remember: A vector has both **magnitude** (length) and **direction** — it is an arrow from the origin to a point, not just the point itself.



$\mathbf{v}$ as an arrow on the coordinate grid

Vector Addition

Definition.

For $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

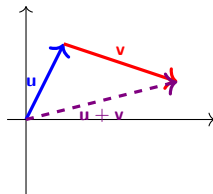
Geometrically: place the tail of $\mathbf{v}$ at the head of $\mathbf{u}$; the sum is the arrow from the start of $\mathbf{u}$ to the end of $\mathbf{v}$ (*tip-to-tail rule*).

Example.

$$\mathbf{u} = (1, 2), \mathbf{v} = (3, -1) \Rightarrow \mathbf{u} + \mathbf{v} = (4, 1).$$

In Plain Words: Doing "walk 1 East, 2 North" then "walk 3 East, 1 South" is the same as one combined instruction: "walk 4 East, 1 North."

Key Idea to Remember: Vector addition is just adding matching coordinates — the operation that makes $(\mathbb{R}^n, +)$ a group (Slide 3).



Tip-to-tail addition

Scalar Multiplication

Definition.

For a scalar $c \in \mathbb{R}$ and vector $\mathbf{v} = (v_1, \dots, v_n)$:

$$c\mathbf{v} = (cv_1, cv_2, \dots, cv_n)$$

It scales the length of $\mathbf{v}$ by $|c|$; if $c < 0$, the direction reverses.

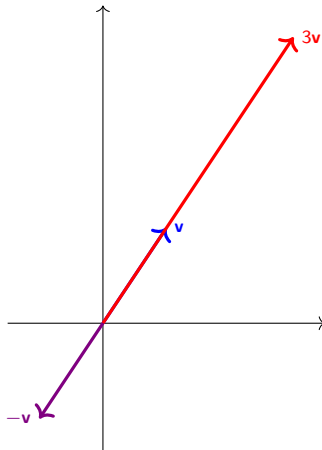
Example.

$\mathbf{v} = (2, 3)$, $c = 3 \Rightarrow 3\mathbf{v} = (6, 9)$ (same direction, $3\times$ longer).

$c = -1 \Rightarrow -\mathbf{v} = (-2, -3)$ (opposite direction, same length).

In Plain Words: Scalar multiplication is a zoom slider: turn it up and the arrow stretches; make it negative and the arrow flips backward.

Key Idea to Remember: Addition + scalar multiplication together are called **linear operations** — nearly every idea in this module builds from just these two moves.



Scaling stretches; a negative sign flips direction

Linear Combinations and Span

Definition.

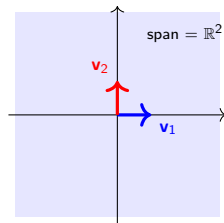
A **linear combination** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k$ for scalars c_j . The **span** of $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of *all* possible linear combinations of them.

Example.

$\text{span}\{(1, 0), (0, 1)\} = \mathbb{R}^2$ (every point reachable). $\text{span}\{(1, 2)\}$ is just the line through the origin with slope 2.

In Plain Words: With only "walk East" and "walk North" (any amount, even negative), the span is every location reachable using just those two tools.

Key Idea to Remember: Span answers: *"What is the largest region I can reach using only these vectors?"*



Two independent vectors span the whole plane

Basis Vectors

Definition.

A set $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a **basis** of a space V if:

- ① it **spans** V (reaches everywhere in V), and
- ② it is **linearly independent** (no vector in the set is a combination of the others — none is "wasted").

Example.

$\{(1, 0), (0, 1)\}$ is the **standard basis** of $\mathbb{R}^2$: minimal, spans everything. $\{(1, 0), (0, 1), (1, 1)\}$ spans $\mathbb{R}^2$ too, but is **not** a basis — it has one vector too many (not independent).

Key Idea to Remember: A basis is the **smallest possible toolkit** of vectors that can still build every vector in the space. The number of vectors in a basis is the **dimension** of the space.

Matrices as Linear Transformations

Definition.

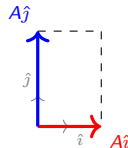
A matrix A is a rectangular array of numbers. Multiplying Ax takes a vector x and moves it to a new vector — this is called a **linear transformation**. The columns of A tell you exactly where the basis vectors land.

Example.

$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ sends $(1, 0) \mapsto (2, 0)$ and $(0, 1) \mapsto (0, 3)$: stretches the plane $2\times$ horizontally, $3\times$ vertically.

In Plain Words: A matrix is a recipe card: "wherever the basis arrows used to point, here is where they point now."

Key Idea to Remember: Ax is a **picture-moving machine** that keeps grid lines straight, evenly spaced, and passing through the origin.



Basis vectors land at the matrix's columns

Matrix Multiplication as Composition

Definition.

If A and B are matrices, the product AB represents applying transformation B **first**, then A : $(AB)\mathbf{x} = A(B\mathbf{x})$.

Example.

Let B rotate the plane 90° , and A stretch it $2\times$. Then $AB\mathbf{x}$ means: first rotate, then stretch. In general $AB \neq BA$ — order matters!

In Plain Words: Think of two instructions on a robot arm: "rotate, then stretch" usually gives a different final pose than "stretch, then rotate." Matrix multiplication records exactly this order.

Key Idea to Remember: Matrix multiplication = doing one transformation right after another. This is why matrix multiplication is **associative** $(AB)C = A(BC)$ but generally **not commutative** $AB \neq BA$.

Three-Dimensional Linear Transformations

Definition.

A linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is represented by a 3×3 matrix. Each column shows where $\hat{i} = (1, 0, 0)$, $\hat{j} = (0, 1, 0)$, $\hat{k} = (0, 0, 1)$ land after the transformation.

Example.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

stretches only the z -direction by $2\times$, leaving the xy -plane untouched — like squashing or stretching a block of clay along one axis.

In Plain Words: Everything that worked for a flat 2D grid (Slide 8) works the same way in 3D — you just now have three basis directions to track instead of two, so the "recipe card" matrix has three columns.

Key Idea to Remember: Extending from 2D to 3D changes *how much* bookkeeping you do, but not the *idea*: a matrix always tells you where the basis vectors go.

The Determinant

Definition (2×2).

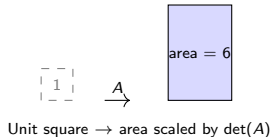
For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(A) = ad - bc$. Measures the **scaling factor of area** (2D) or **volume** (3D) caused by A .

Example.

$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \Rightarrow \det(A) = 6$: a unit square becomes a rectangle with $6\times$ the area. If $\det(A) = 0$, space is squashed into a lower dimension — all area/volume destroyed.

In Plain Words: Photocopying a picture $2\times$ wider and $3\times$ taller covers $6\times$ the paper. The determinant is exactly this "area multiplier," negative if the map also flips (mirrors) space.

Key Idea to Remember: $\det(A) = 0 \iff$ columns of A are dependent
 $\iff A$ is **not invertible**.



Inverse Matrices

Definition.

The **inverse** A^{-1} of a square matrix A satisfies $AA^{-1} = A^{-1}A = I$ (the identity matrix). It exists **iff** $\det(A) \neq 0$. Solving $A\mathbf{x} = \mathbf{b}$ becomes $\mathbf{x} = A^{-1}\mathbf{b}$.

Example.

$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \Rightarrow A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}$ — it exactly **undoes** the stretching done by A .

In Plain Words: If A is a transformation that "stretches $2\times$ sideways, $3\times$ up", then A^{-1} is the "un-stretch" that shrinks everything back to exactly where it started.

Column Space and Null Space

Definition.

Column space $\text{Col}(A)$: the span of A 's columns — i.e., every possible output $A\mathbf{x}$ can land in. **Null space** $\text{Null}(A)$: all $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$ — the vectors that get squashed to the origin.

Example.

$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: both columns point the same direction, so $\text{Col}(A)$ is just a line, not the whole plane. Its null space is the line $x_1 = -2x_2$ — every point on that line is crushed to $(0, 0)$.

Key Idea to Remember: Rank–Nullity Theorem: $\dim(\text{Col}(A)) + \dim(\text{Null}(A)) = n$ (number of columns). Column space measures "where you can land"; null space measures "what gets lost."

Nonsquare Matrices as Transformations Between Dimensions

Definition.

An $m \times n$ matrix ($m \neq n$) represents a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ — it changes the number of dimensions, not just the shape of the space.

Example.

$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ is 3×2 : it takes a 2D input and produces a 3D output — it "lifts" a flat plane into 3D space (input dimension $2 <$ output dimension 3). A 2×3 matrix instead **flattens** 3D space onto a 2D plane, like a shadow.

Key Idea to Remember: A matrix does not need to be square! Rows = dimension of the output space, columns = dimension of the input space. Rectangular matrices move information **between** different-sized spaces.

Dot Product

Definition.

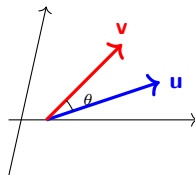
$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$, where θ is the angle between them.

Example.

$\mathbf{u} = (1, 2)$, $\mathbf{v} = (3, -1) \Rightarrow \mathbf{u} \cdot \mathbf{v} = 1(3) + 2(-1) = 1$. If

$\mathbf{u} \cdot \mathbf{v} = 0$, the vectors are **perpendicular**.

In Plain Words: The dot product answers: "how much does one arrow point in the same direction as another?" Big positive = same way; zero = right angles; negative = somewhat opposite.



Angle θ between $\mathbf{u}$ and $\mathbf{v}$

Duality

Definition.

Every linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}$ (a "linear functional") can be written as a dot product with some fixed vector $\mathbf{u}$: $T(\mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$. This $\mathbf{u}$ is the **dual vector** of T .

Example.

The 1×2 matrix $(2 \ 3)$ acting on (x, y) gives $2x + 3y$ — identical to $(2, 3) \cdot (x, y)$. The "row matrix" and the "vector" are secretly the same object viewed two ways.

Key Idea to Remember: Duality is the surprising fact that "linear transformations that output a single number" and "vectors" are two sides of the same coin — every such transformation *is* a dot product with a specific vector.

Hyperplanes

Definition.

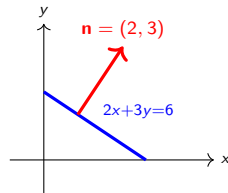
A **hyperplane** in $\mathbb{R}^n$ is a flat subset of dimension $n - 1$: the solution set of one linear equation $\mathbf{n} \cdot \mathbf{x} = c$, $\mathbf{n} \neq \mathbf{0}$, where $\mathbf{n}$ is the **normal vector** (perpendicular to it) and c a constant.

Example.

In $\mathbb{R}^2$, a hyperplane is a **line**: $2x + 3y = 6$ has normal $\mathbf{n} = (2, 3)$. In $\mathbb{R}^3$, it is an ordinary **plane**: $x + y + z = 1$.

In Plain Words: A point cuts a line, a line cuts a plane, a plane cuts 3D space — the normal vector is a flagpole sticking straight out, showing which way is "up" from it.

Key Idea to Remember: A hyperplane through the origin is exactly the **null space** (Slide 9) of $\mathbf{x} \mapsto \mathbf{n} \cdot \mathbf{x}$ — also the key object behind decision boundaries in ML classifiers.



Line = hyperplane in $\mathbb{R}^2$; $\mathbf{n}$ perpendicular to it

Cross Product

Definition.

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$:

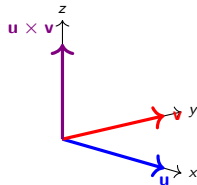
$$\mathbf{u} \times \mathbf{v} = (u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1)$$

A new vector **perpendicular to both $\mathbf{u}$ and $\mathbf{v}$** , with length $\|\mathbf{u}\| \|\mathbf{v}\| \sin \theta = \text{area}$ of the parallelogram they span.

Example.

$\mathbf{u} = (1, 0, 0)$, $\mathbf{v} = (0, 1, 0) \Rightarrow \mathbf{u} \times \mathbf{v} = (0, 0, 1)$ — perpendicular to the xy -plane, pointing "up," by the right-hand rule.

In Plain Words: Only in 3D does "perpendicular to two arrows" pin a unique direction. Point your right-hand fingers from $\mathbf{u}$ to $\mathbf{v}$; your thumb shows $\mathbf{u} \times \mathbf{v}$.



Right-hand rule: perpendicular to $\mathbf{u}$ and $\mathbf{v}$

Cross Products in the Light of Linear Transformations

Definition.

$\mathbf{u} \times \mathbf{v}$ can be defined via a determinant / dual-vector trick: define the linear functional $T(\mathbf{x}) = \det[\mathbf{x} \ \mathbf{u} \ \mathbf{v}]$ (a 3×3 determinant with $\mathbf{x}$ as first column). By duality (Slide 11), $T(\mathbf{x}) = \mathbf{p} \cdot \mathbf{x}$ for a unique $\mathbf{p}$ — and that $\mathbf{p}$ is exactly $\mathbf{u} \times \mathbf{v}$.

Key Idea to Remember: This reframes the cross product as: *"the vector whose dot product with anything equals the signed volume of the parallelepiped formed with $\mathbf{u}$ and $\mathbf{v}$."* It elegantly ties together determinants (Slide 8), duality (Slide 11), and the cross product into one idea.

In Plain Words: Instead of memorising the messy cross-product formula, remember: it is "the special vector that reports volumes" — built from the determinant, discovered using the duality trick.

Cramer's Rule

Definition.

For $Ax = \mathbf{b}$ with $\det(A) \neq 0$, each unknown is:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is A with column i replaced by $\mathbf{b}$.

Example.

$\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases}$, $A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$, $\det(A) = -3$. Replacing column 1 with $(5, 1)$: $\det(A_1) = -6 \Rightarrow x = 2$. Replacing column 2: $\det(A_2) = -3 \Rightarrow y = 1$.

In Plain Words: Cramer's Rule uses the determinant's "area/volume" meaning (Slide 8): swapping in $\mathbf{b}$ and comparing the new volume to the old volume directly tells you each unknown's value — no elimination steps needed.

Key Idea to Remember: Cramer's Rule is elegant for small systems and for theory, but computationally expensive for large n ; Gaussian elimination is faster in practice.

Change of Basis

Definition.

If $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis, any vector $\mathbf{v}$ can be written in B -coordinates. The **change-of-basis matrix** P (whose columns are the $\mathbf{b}_i$, written in the standard basis) converts between the two:

$$\mathbf{v}_{\text{standard}} = P \mathbf{v}_B, \quad \mathbf{v}_B = P^{-1} \mathbf{v}_{\text{standard}}$$

Example.

New basis $\mathbf{b}_1 = (2, 1)$, $\mathbf{b}_2 = (-1, 1)$. A vector that is "(1, 1) in the new basis" corresponds to $1 \cdot \mathbf{b}_1 + 1 \cdot \mathbf{b}_2 = (1, 2)$ in the standard coordinates.

In Plain Words: Changing basis is like converting temperature between Celsius and Fahrenheit — same physical reality (the same arrow), described using a different scale/grid.

Key Idea to Remember: To translate a transformation matrix A (given in the standard basis) into the new basis B , use $A_B = P^{-1}AP$.

Eigenvectors and Eigenvalues

Definition.

For a square matrix A , a nonzero vector $\mathbf{v}$ is an **eigenvector** with **eigenvalue** λ if:

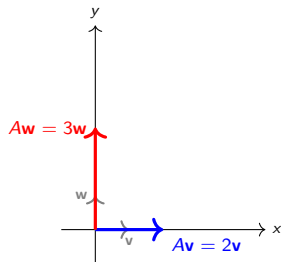
$$A\mathbf{v} = \lambda\mathbf{v}$$

A only **stretches/shrinks** $\mathbf{v}$ (by factor λ) — it does **not** rotate it off its own line. Found by solving $\det(A - \lambda I) = 0$ (the **characteristic equation**).

Example.

$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$: eigenvectors $(1, 0)$ with $\lambda = 2$, and $(0, 1)$ with $\lambda = 3$ — the axes don't rotate, only stretch.

In Plain Words: Most vectors get knocked off their line by a transformation. Eigenvectors are the rare directions that stay on the **same line**, just longer or shorter — like the axis a spinning top wobbles around.



Eigenvectors stay on their own line, only scaled

Diagonalization

Key Idea to Remember: If a matrix A has enough independent eigenvectors, we can write $A = PDP^{-1}$, where D is diagonal (eigenvalues on the diagonal) and P 's columns are the eigenvectors. This is exactly a **change of basis** (Slide 16) into "eigenvector coordinates," where the messy transformation A becomes simple scaling D .

Why it matters.

Diagonalization makes repeated transformations trivial: $A^k = PD^kP^{-1}$, and D^k is just each diagonal entry raised to the power k — used constantly in algorithms (e.g., PCA, Markov chains, PageRank).

Vector Spaces — Axioms

Definition.

A **vector space** V over a field F (Slide 4) is a set with addition and scalar multiplication satisfying 8 axioms, including:

- $(V, +)$ is an abelian group (closure, associativity, identity $\mathbf{0}$, inverses)
- $1 \cdot \mathbf{v} = \mathbf{v}$; $(ab)\mathbf{v} = a(b\mathbf{v})$
- Distributivity: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$, $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

Example.

$\mathbb{R}^n$ is the classic vector space. Less obvious examples: the set of all polynomials of degree ≤ 2 , or the set of all 2×2 matrices — both satisfy every axiom, so both are legitimate "vector spaces," even though their elements are not arrows.

Key Idea to Remember: Once a system obeys these 8 rules, **everything** we built in this module — span, basis, linear transformations, eigenvectors — automatically applies to it, whether the "vectors" are arrows, polynomials, or matrices.

Subspaces

Definition.

A **subspace** $W \subseteq V$ is a subset that is itself a vector space: it must contain $\mathbf{0}$, and be closed under addition and scalar multiplication.

Example.

Any line or plane through the origin in $\mathbb{R}^3$ is a subspace. The column space and null space (Slide 9) are both subspaces — one of the output space, one of the input space.

In Plain Words: A subspace is a "space within a space" that never lets you leave through addition or scaling — it must include the origin, since $0 \cdot \mathbf{v} = \mathbf{0}$ always.

Mathematical Spaces: The Bigger Picture

In Plain Words: A vector space (Slide 18) tells you how to add and scale. But it says nothing about "distance" or "angle" between two elements. **Mathematical spaces** add extra structure on top of a set to capture ideas like closeness, length, and angle.

Definition: Metric Space.

A **metric space** (X, d) is a set X with a distance function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ satisfying:
 $d(x, y) = 0 \iff x = y$; symmetry $d(x, y) = d(y, x)$; and the triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.

Example.

$\mathbb{R}^n$ with $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ (ordinary distance) is a metric space. So is a city grid using "taxicab distance"
 $|x_1 - y_1| + |x_2 - y_2|$ — a different, but still valid, notion of distance.

Normed and Inner Product Spaces

Definition: Normed Space.

A **norm** $\|\cdot\|$ assigns a non-negative "length" to each vector, satisfying $\|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$, $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$, and the triangle inequality. A vector space with a norm is a **normed space**; every norm automatically gives a metric via $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Definition: Inner Product Space.

An **inner product** $\langle \cdot, \cdot \rangle$ generalises the dot product (Slide 11) to any vector space, satisfying linearity, symmetry, and positive-definiteness. It gives both a norm ($\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$) and a notion of **angle**.

Key Idea to Remember: The chain of generality is: **Inner Product Space** $\Rightarrow$ **Normed Space** $\Rightarrow$ **Metric Space**. Each step keeps less structure: angle and length $\rightarrow$ only length $\rightarrow$ only distance. $\mathbb{R}^n$ with the dot product sits at the richest end of this chain — which is exactly why it is the space used throughout this module.

Module 2: Summary

- **Algebraic structures** (group/ring/field) give the rules of arithmetic that vectors and scalars must obey.
- **Vectors** live in a **coordinate system**; **addition** and **scalar multiplication** are the two basic legal moves.
- **Span** and **basis** describe what region vectors can reach, and the minimal toolkit to reach it.
- **Matrices** are transformations; **multiplication** is composing them; this extends unchanged from 2D to 3D and to **nonsquare** maps between dimensions.
- **Determinant** measures area/volume scaling and invertibility; **inverse**, **column space**, and **null space** describe what a matrix can undo, reach, and destroy.
- **Dot product** and **duality** connect vectors to linear functionals; **hyperplanes** are the level sets of those functionals.
- **Cross product** builds a perpendicular vector using exactly this duality trick with determinants.
- **Cramer's Rule** solves systems directly via determinants; **change of basis** re-expresses everything in a new coordinate grid.
- **Eigenvectors/eigenvalues** find the directions a transformation merely scales; **vector spaces** and **mathematical spaces** generalise all of the above beyond arrows in $\mathbb{R}^n$.

End of Module 2 Notes